AP PRECALCULUS

Unit 1 • Worksheet 1.1-1.6B • Answer key

Review 1.1-1.6B • Review of Topics 1.1 to 1.6

Answer key with worked solutions

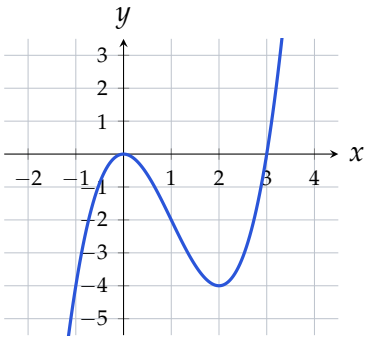
Directions

Answer every question. Multiple-choice questions have one best answer. On free-response questions, show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

Question 1 (1.1 • graph)

NO CALCULATOR

The graph of f is shown. On which interval is f increasing and concave up?



(A) $(0, 1)$
(B) $(1, 2)$
(C) $(2, \infty)$
(D) $(-\infty, 0)$

Solution

Answer: (C). $f(x) = x^2(x-3)$ increases on $(-\infty, 0)$ and $(2, \infty)$; concave up for $x > 1$.

Question 2 (1.2 • graph)

NO CALCULATOR

Using the same graph, what is the average rate of change of f on $[1, 3]$?

(A) -2
(B) 0
(C) 2
(D) 1

Solution

Answer: (D). $f(1) = -2$, $f(3) = 0$: $\frac{2}{2} = 1$.

Question 3 (1.5) NO CALCULATOR

Which could be the polynomial graphed above?

(A) $x(x-3)^2$
(B) $-x^2(x-3)$
(C) $x^2(x-3)$
(D) $x^2(x+3)$

Solution
Answer: (C). Touches at 0, crosses at 3, rises to the right.

Question 4 (1.3) NO CALCULATOR

For $g(x) = -2x^2 + 8x - 1$, on which interval is g increasing?

(A) $(-\infty, 2)$
(B) $(2, \infty)$
(C) $(-\infty, -2)$
(D) $(-2, \infty)$

Solution
Answer: (A). Vertex $x = -\frac{8}{-4} = 2$; $a < 0$, so increasing before the vertex.

Question 5 (Free response) NO CALCULATOR

The table gives values of a function h whose graph has no breaks.

x	-3	-2	-1	0	1	2	3
$h(x)$	-10	0	2	2	2	0	-10

(a) Find the average rate of change on $[-3, -1]$ and on $[1, 3]$.
(b) Is h more likely even or odd? Explain.
(c) The values come from a polynomial of degree 4. Give its end behavior.

Solution
(a) $\frac{2-(-10)}{2} = 6$; $\frac{-10-2}{2} = -6$.
(b) $h(-x) = h(x)$ for every value shown: even.
(c) Both ends go down ($\lim_{x \to \pm\infty} h(x) = -\infty$) since h falls at both ends of the table.